

Data Visualization: Project 2

Introduction – Objectives, Motivation and Problem

Artificial intelligence has become a strategic asset with economic, technological, and geopolitical impact. This project analyzes Spain's position within the global AI ecosystem, identifying its strengths and weaknesses, and comparing it with similar countries in order to propose an optimal investment strategy based on data from the Global AI Index.

The main objectives are:

- Identify the leading countries and understand their regional and income-level distribution.
- Evaluate investment across the main pillars (Talent, Infrastructure, R&D, Operating Environment, Strategy, and Commercial) and position Spain within this framework.
- Detect gaps between governmental strategy and actual R&D capacity.
- Analyze the effect of political regime on AI performance.
- Compare Spain with 'mirror countries' similar in economic size and demographics.
- Measure the quantitative gap between Spain and benchmark countries.

The motivation is to provide a useful guide for public decision-makers: not only to identify who leads in AI, but also to determine what Spain should prioritize given resource constraints and the varying levels of maturity across pillars. The project is grounded in common structural challenges: insufficient or poorly allocated investment, lack of sustained institutional support, weaknesses in infrastructure, and underdeveloped innovation ecosystems.

Related Work – Inspiring Visualizations or Articles

The project is based on the methodology of Tortoise Media's Global AI Index, which relies on 122 indicators and seven sub-pillars that structure the analysis. Although the version used here does not exactly match the 2023 documentation, it serves as a conceptual reference.

The sources that inspire the design of our visualizations include

- **Global AI Index (Tortoise):** maps, rankings, and country profiles; provides the conceptual pillars and encodings such as the use of flags.
- **Intersog – Global AI Race: Dominant Players and Aspiring Challengers:** introduces the idea of distinguishing between leaders, challengers, and countries with potential.

Our project differs because:

- It focuses on Spain as a case study, in addition to providing a global perspective.
- It introduces derived metrics (Strategy Gap, Intensity, Global_AI) to evaluate strategic coherence and efficiency.
- It adds a comparison with economically similar countries, developing the concept of "mirror countries" beyond the general ranking.

4. Data – Dataset Description and Preparation

4.1 Dataset Source and Structure

The dataset used, `AI_index_db.csv`, is a tabular version of the Global AI Index adapted to the context of this project. It contains 62 countries and, after cleaning, includes the following main variables:

- **Country:** name of the country.

- **Talent, Infrastructure, Operating Environment, Research, Development, Government Strategy, Commercial:** the seven sub-pillars of the index, already scaled to a 0–100 range.
- **Total score:** overall AI capability score (0–100), aggregating the seven sub-pillars according to Tortoise's methodology
- **Region:** geographic macro-region (Americas, Europe, Asia-Pacific, Middle East, Africa).
- **Cluster:** additional groupings defined by the index → not used in the visualizations.
- **Income group:** country income group (High, Upper middle, Lower middle).
- **Political regime:** political regime category (Liberal Democracy, Closed Autocracy, Electoral Democracy, Electoral Autocracy), used to study the relationship between regime and Total Score

Additionally, for more advanced analyses, economic variables and country codes are incorporated:

- **GDP_2023:** 2023 GDP in current dollars
- **POP_2023:** 2023 population
- **iso3 / iso:** country ISO codes, used to link with other resources and to construct flag URLs.
- **flag_url:** URL of each country's flag (for enhanced charts with icons).

On the other hand, in this project we have applied several classic principles of data visualization design:

- Selection of chart type according to the question (maps for "where," bars for "who," radars for profiles).
- Correct use of perceptual encodings (position and length for comparisons, sequential color for intensity, divergent color for differences)
- Normalized data and derived metrics to ensure comparability.
- Visual consistency (single template) and small multiples to avoid overload.
- Interactivity to display details on demand.

4.2 Cleaning and Preprocessing

- Correction of names and specific adjustments to GDP/population when they were missing or inconsistent.
- Filtering of countries without valid economic data to avoid errors in derived metrics.
- Enrichment with ISO codes and flag URLs for interactive visualizations.

4.3 Derived Variables

Several calculated variables have been defined to answer the analytical questions:

- **Strategy Gap** A gap analysis is a technique that compares the current state of an organization with the desired state, identifying the distance between the two and helping to design strategies to close it.

$$\text{Strategy Gap} = \text{Government Strategy} - \frac{\text{Research} + \text{Development}}{2}$$

This metric attempts to capture whether a country's government promises more than its R&D capacity can support (large positive gaps) or if, instead, its policies are aligned with its real performance.

Intensity: To approximate the **intensity of AI investment relative to the country's size**, a metric is constructed that combines the Total score with GDP and population (AI "per million inhabitants and per trillion GDP"). Conceptually, this variable indicates whether a country is "punching above its weight" in AI given its economic and demographic size.

$$\text{Intensity} = \frac{\text{Population_norm}}{2} + \frac{\text{PIB_norm}}{2}$$

Global_AI: The analysis of countries comparable to Spain, a composite metric is created:

$$\text{Global_AI} = 0,7 \cdot \text{Total_norm} + 0,3 \cdot \text{Intensity_norm}$$

where Total_norm and Intensity_norm are standardised versions within the group of countries with GDP and population between half and twice that of Spain. This metric gives more weight to absolute performance (Total score), but it also gives importance to Intensity. This conclusion is achieved since Intensity penalizes large countries and favors small countries a lot, although in the end what really matters to us is those countries with a high impact on AI.

4.4 Specific subsets for the analysis of Spain

To answer the questions focused on Spain, additional subsets were defined:

- **Comparable countries (Q4):** countries with GDP and population between 0.5× and 2× those of Spain; from them the best positioned countries are selected (Canada, South Korea, Australia, France).
- **Bilateral comparisons (Q9):** differences by pillar between Spain and each mirror country using $\Delta = \text{Spain} - \text{País_comparado}$.

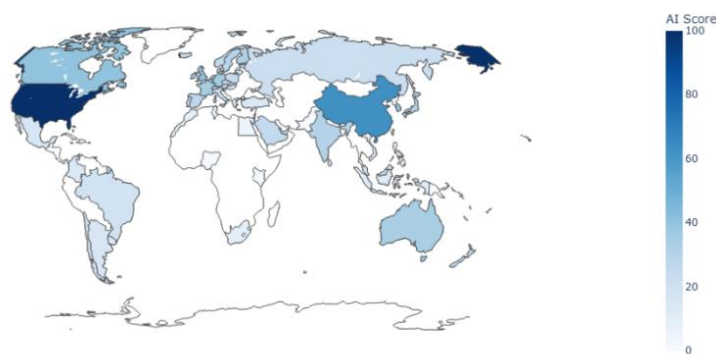
5. Analytical Questions

The analysis is structured around several analytical questions, which guide both the selection of visualizations and the construction of metrics. As we said earlier, we use a structure with ascending order of complexity:

Q1. Which countries have the highest and lowest Total Score values and how are they distributed by region and income group?

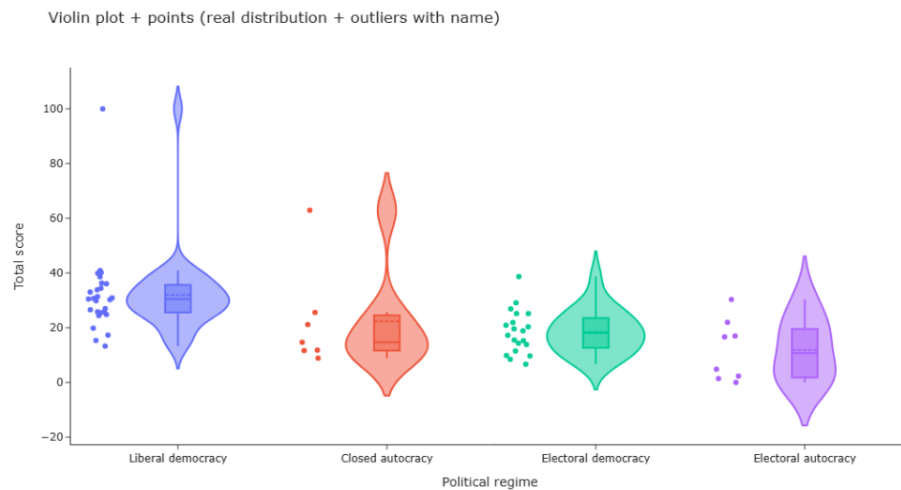
- Show **where** global AI capability is concentrated.
- Allow for the rapid identification of **leaders, laggards, and regional patterns**.
- **Sequential color** favors the perception of continuous graduations (theory of perceptual scales).
- Maps are suitable for *overview tasks first* → seeing global spatial patterns.
- Blue conveys neutrality and works well for quantitative values.

Global AI Index 2025 - Global Leadership in AI



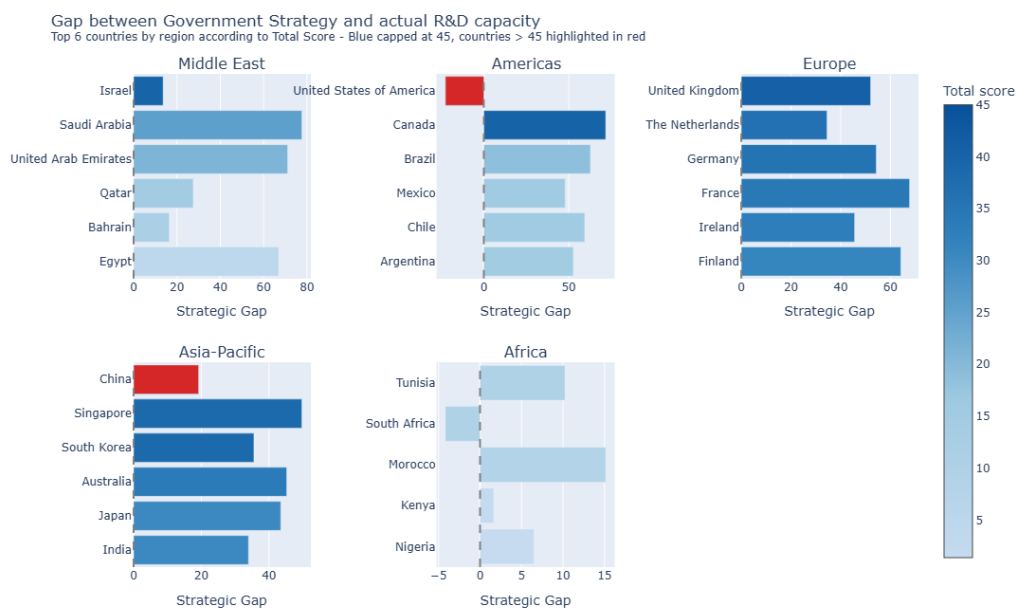
Q6. Does the political regime influence AI?

- Show distribution, variability and outliers by category.
- Violin + points + boxplot superimposed for each type of regime.
- Violins show full densities (form of distribution).
- Boxplots allow you to compare medians and ranges very efficiently.

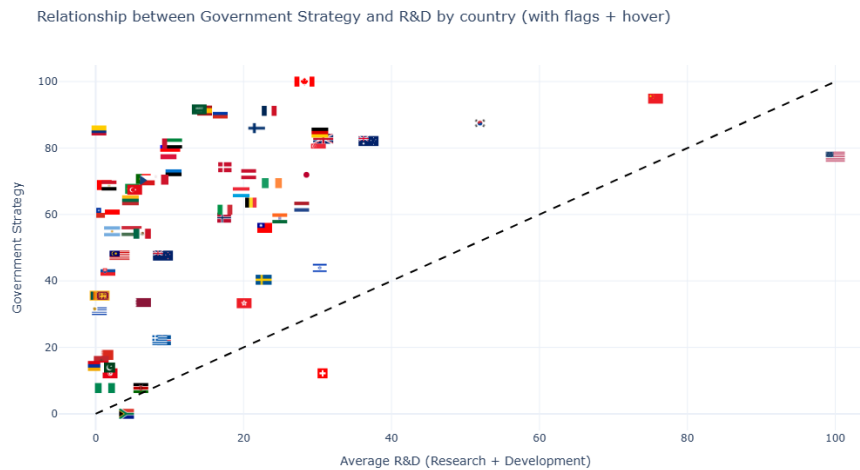


Q3. Which countries have a gap between the Government Strategy and the actual capacity for Research and Development?

- Allow countries to be compared within each region.
- Multiple small (5 subplots per region) for easy viewing without clutter
- Horizontal bars colored by region, opacity proportional to the magnitude of the gap
- Horizontal bars maximize visual accuracy of **comparison by length**
- Different colors in the US and China to highlight their great potential in AI compared to the others



- It allows identifying **misalignments between political strategy and real R+D capacity** in each country.
- Makes it easy to detect **consistent/inconsistent clusters** based on AI maturity level
- Diagonal line as a reference that improves the reading of deviations
- Use of flags → Promote quick identification without the need for textual labels → *visual recognition > reading*

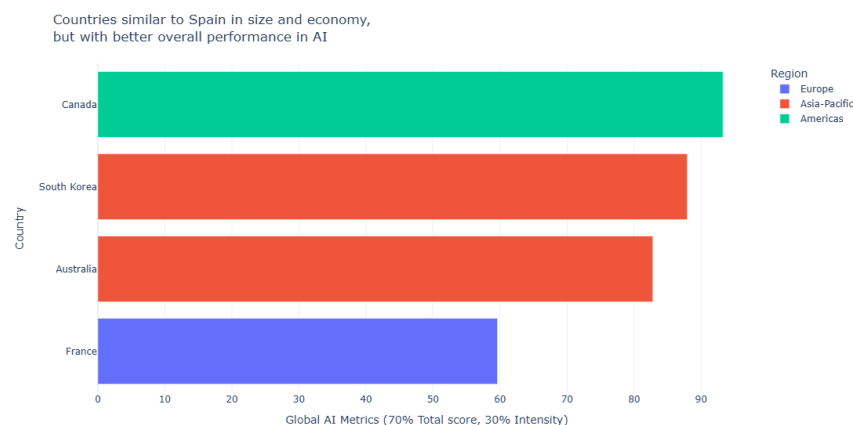


KEY DIFFERENCE Q3: with the first graph we focus on the most relevant countries improving visual quality, while the second provides us with a more generic perspective encompassing all countries.

Q4. Which countries have a better AI investment strategy (Intensity)? Which are the countries with a situation similar to Spain (GDP/population) with the best strategy (Total score)?

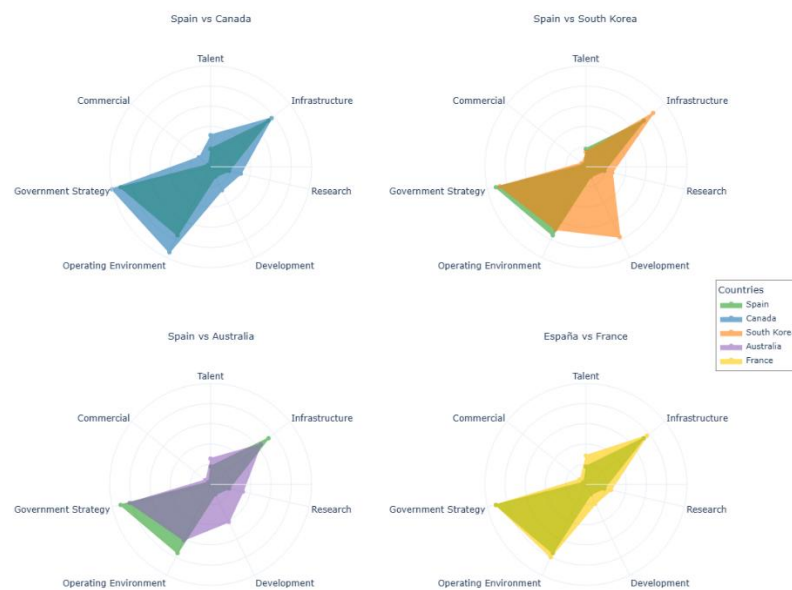
- Identify **countries comparable to Spain** that have achieved better results in AI.
- Provide a composite metric that combines Total Score and Intensity
- Simple but accurate and visual → use of horizontal bars
- **Categorical palette by region** → Reinforces the regional analysis of the project
- Rich Tooltip → Allows detailed inspection without cluttering the graph
- Descending order → Minimizes cognitive load and directs attention to the most relevant countries

KEY DECISION IN IMPLEMENTATION: We normalized Total Score to [0, 100] again because now we are not interested in the world ranking, but in the ranking within the group similar to Spain.



Q2. What common investment patterns stand out in these four countries with the highest AI performance and how do they compare to Spain's current strategy?

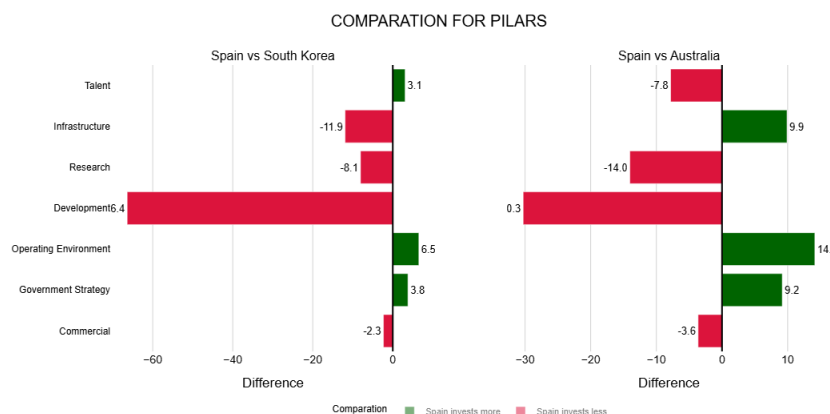
- Compare multivariate profiles across countries across the seven pillars of AI.
- Identify **Spain's structural strengths and weaknesses**.
- Radars allow you to detect **shapes and patterns** (min/max) intuitively.
- Transparency allows for **perceptible overlap** without losing readability.
- Ideal for **profiling** tasks, not fine comparison (complemented by → Q9 differential bars)



Q9. In which pillars of investment in AI does Spain differ from South Korea and Australia? What is the magnitude of this gap?

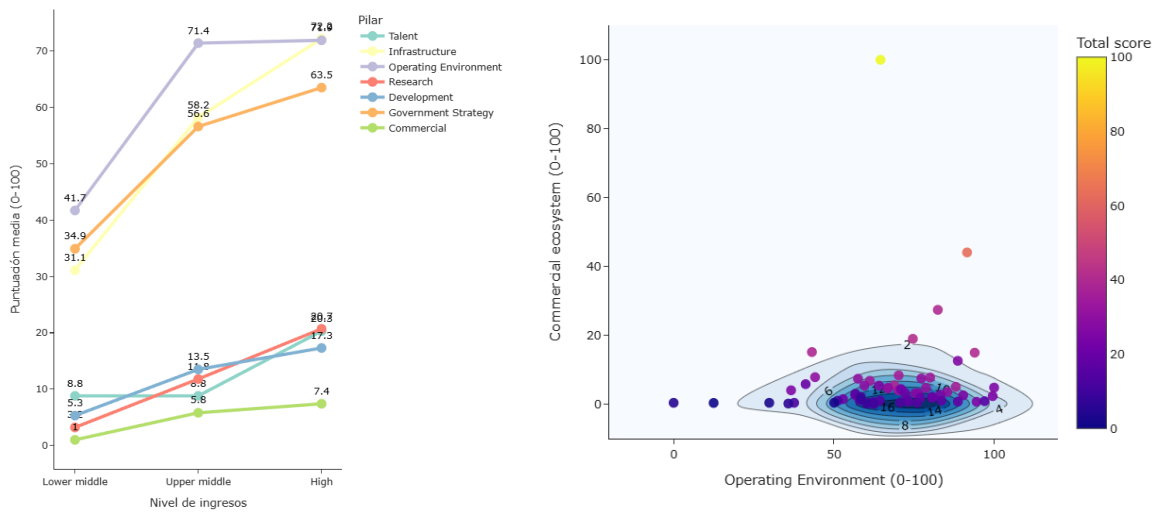
- Quantitatively measure the **gap by pillar** between Spain and another country
- Two horizontal bar subplots centered on 0
- Divergent Colors and Vertical Reference Line
- The use of **divergent color** (green/red) is optimal for encoding signed differences.
- The relative position with respect to **0** facilitates quick judgments of magnitude

KEY DECISION IN THE IMPLEMENTATION: In Plotly the legend works by trace, not by color → as we have two subplots so we hide the two real traces of the legend (showlegend=False) and add two "dummy" traces only for the legend: one green and one red.



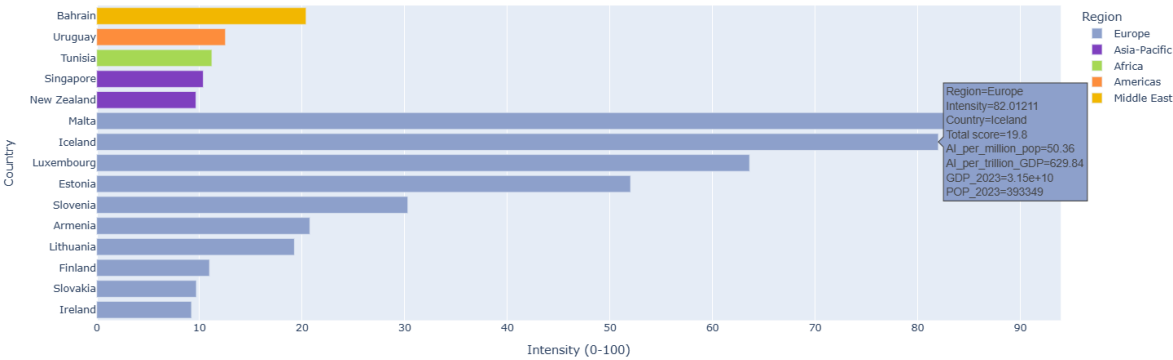
Work development:

Throughout the project, we initially started with a broad set of Analytical Questions and several visualization alternatives for each one. However, after an iterative evaluation process, we selected only the questions that offered the greatest analytical value, together with the most appropriate and informative visual representations. We then reorganized all visualizations to construct a coherent and progressive visual narrative, structured according to the logical sequence of the analysis (Q1, Q6, Q3, ...). Examples of some visualization alternatives:



There was also an evolution of each graphic, refining the visual encodings until they were in the best possible condition.

One of the main challenges encountered during the development was the computation of the “Intensity” metric. When comparing it with the results reported in the reference article, we identified significant inconsistencies in the distribution obtained. This led us to design a new composite metric, “Global_AI”, which properly weights the *Total Score* and the performance relative to economic scale. This metric provided results that were much more coherent, interpretable, and aligned with the true objectives of our study.



Critical Validation of the Data Visualization Project on the Global AI Ecosystem

The project presents a **Successful Validation** across its four design levels, demonstrating a robust "Problem-driven" approach and effective technical execution.

Nested Model Level	Threat	Key Strengths
1. Domain (Problem)	Misunderstanding the need	The goal of guiding Spain's public AI investment is a real and well-defined business need.

<u>Nested Model Level</u>	<u>Threat</u>	<u>Key Strengths</u>
<u>2. Abstraction</u> (Data & Tasks)	<u>Showing the wrong information</u>	Original and relevant derived metrics were created, such as Strategy Gap and the weighted metric Global_AI , to quantify strategic coherence and efficiency. The selection of "mirror countries" (similar in GDP/population) is key for fair comparative analysis.
<u>3. Idiom</u> (Visualization & Interaction)	<u>Using an ineffective representation</u>	Optimal idioms were selected for the task: Choropleth Maps (overview) Scatter plots with flags (Strategy vs. R&D relationship) , and especially the Diverging Bar Chart (to measure the magnitude of gaps per pillar).
<u>4. Algorithm</u> (Implementation)	<u>Code is too slow</u>	The implementation on a small dataset using optimized libraries (Pandas/Plotly) is efficient and robust . Specific plotting complexities were addressed with supporting code (e.g., the dummy legend for the diverging bar chart).

Conclusion: The project stands out for its **analytical rigor** (creating new metrics for key questions) and its **visual precision**, offering a clear and actionable diagnostic tool.

BIBLIOGRAPHY

<https://www.tortoisemedia.com/data/global-ai>
<https://asana.com/es/resources/gap-analysis>
<https://data.worldbank.org/indicator/NY.GDP.MKTP.CD> (GDP dataset --> GDP)
<https://data.worldbank.org/indicator/SP.POP.TOTL> (population dataset)
<https://www.iso.org/obp/ui/#search> (ISO international standard)
<https://www.ris.gov.tw/app/portal>
<https://www.imf.org/external/pubs/ft/ar/2023/>